

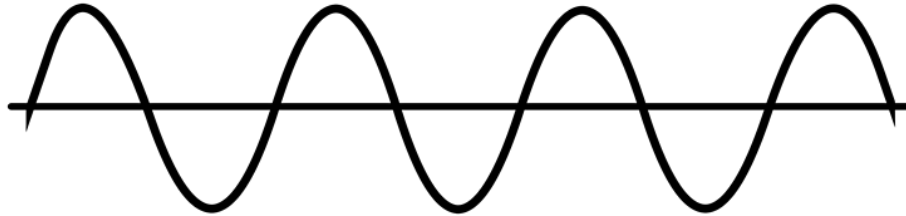
Candidate Voice Pitch Influences Election Outcomes

Casey A. Klofstad, University of Miami

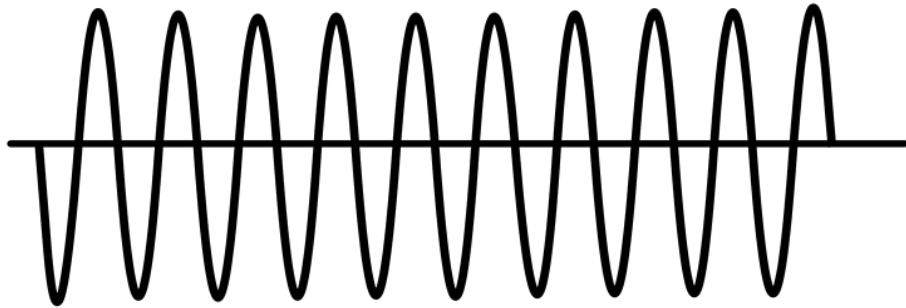
Political Psychology, Vol. 37, No. 5 (OCTOBER 2016), pp. 725-738

Presented by
Paolo Laffranchi

What is (Voice) Pitch?

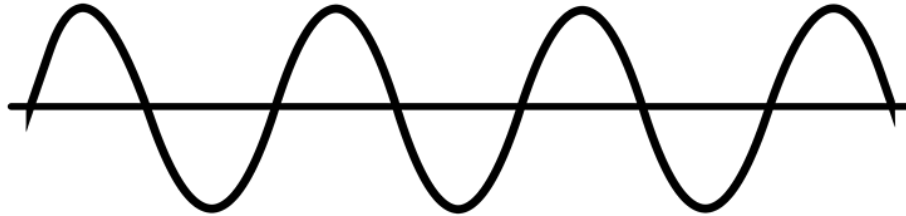


Low-pitched sound

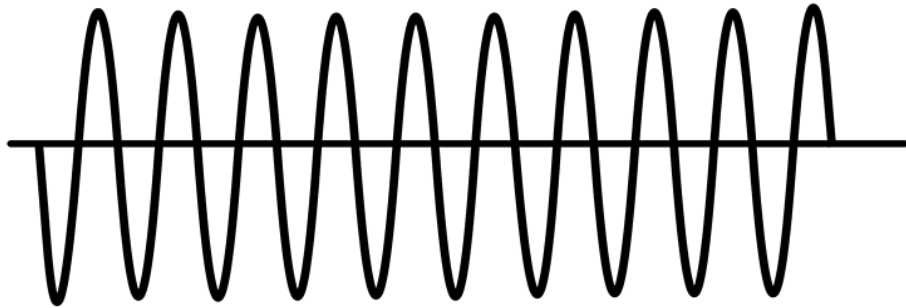


High-pitched sound

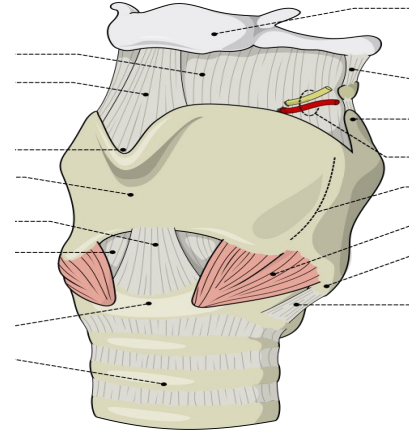
What is (Voice) Pitch?



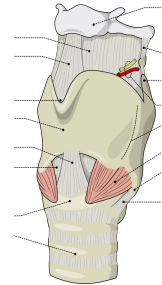
Low-pitched sound



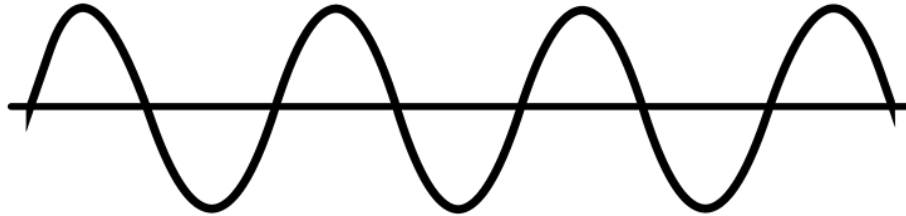
High-pitched sound



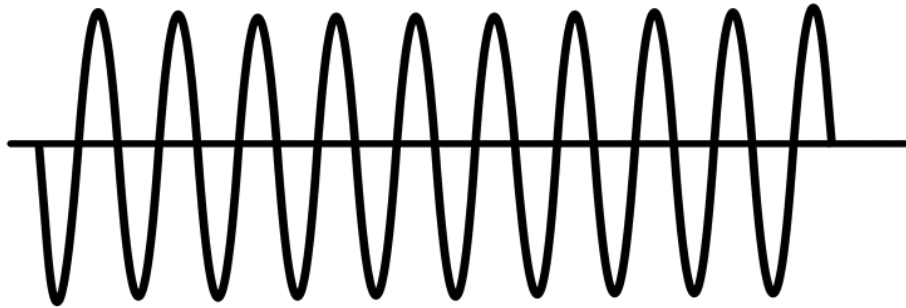
Larynx (vocal-cords)



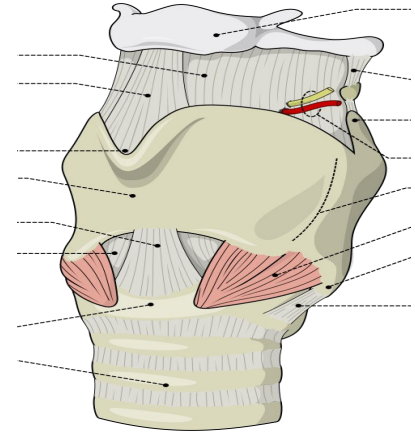
What is (Voice) Pitch?



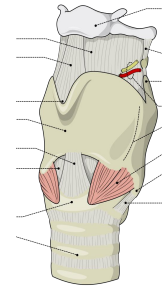
Low-pitched sound



High-pitched sound



Larynx (vocal-cords)



The Importance of Voice Pitch



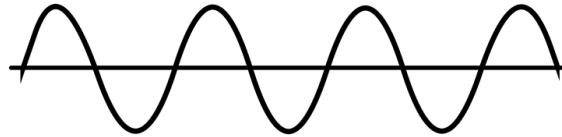
https://commons.wikimedia.org/wiki/File:Roaring_lion_%28Unsplash%29.jpg

The Importance of Voice Pitch

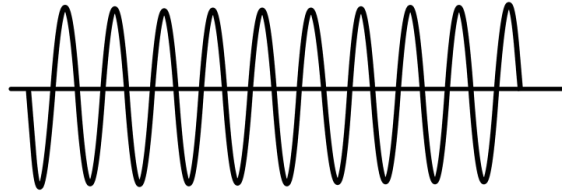
In humans



https://commons.wikimedia.org/wiki/File:Roaring_lion_%28Unsplash%29.jpg



Low-pitched sound



High-pitched sound

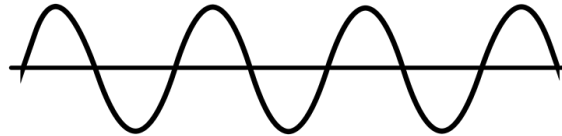


The Importance of Voice Pitch

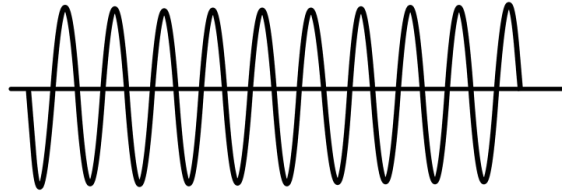
In humans



https://commons.wikimedia.org/wiki/File:Roaring_lion_%28Unsplash%29.jpg



Low-pitched sound



High-pitched sound



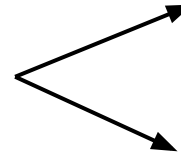
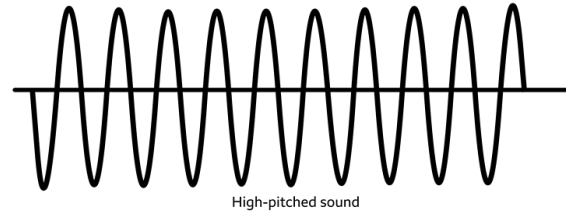
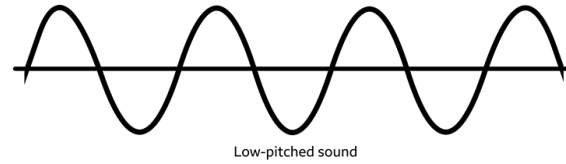
more attractive

The Importance of Voice Pitch

In humans



https://commons.wikimedia.org/wiki/File:Roaring_lion_%28Unsplash%29.jpg

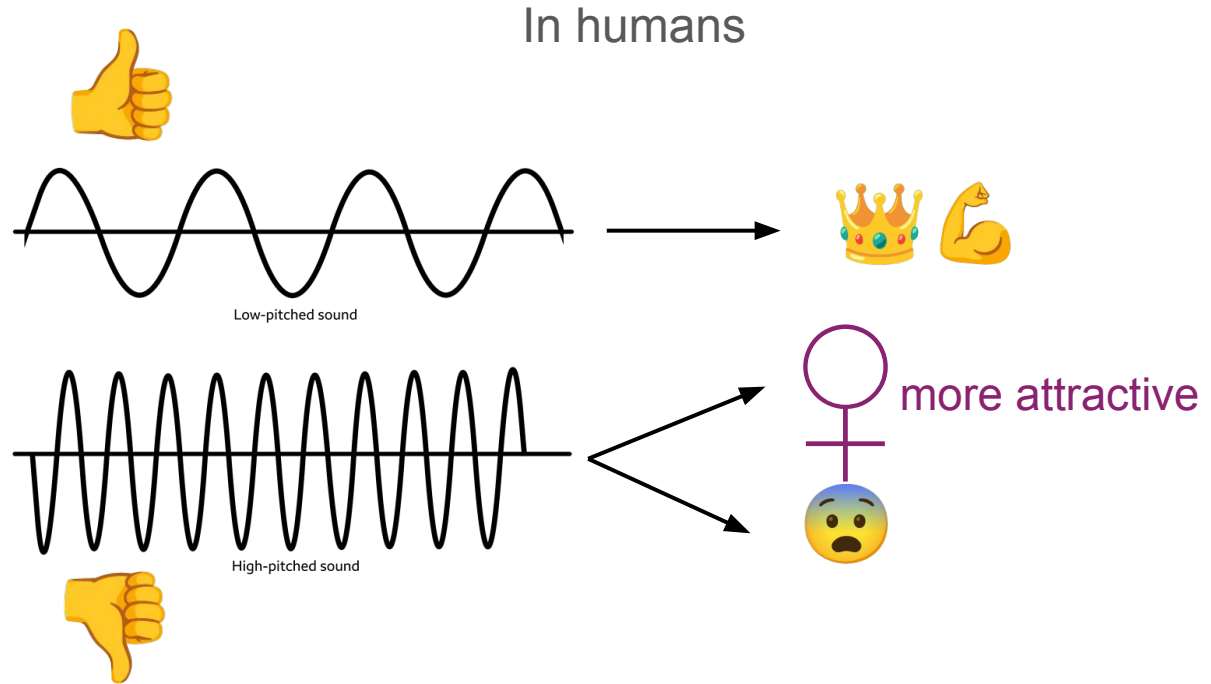


more attractive

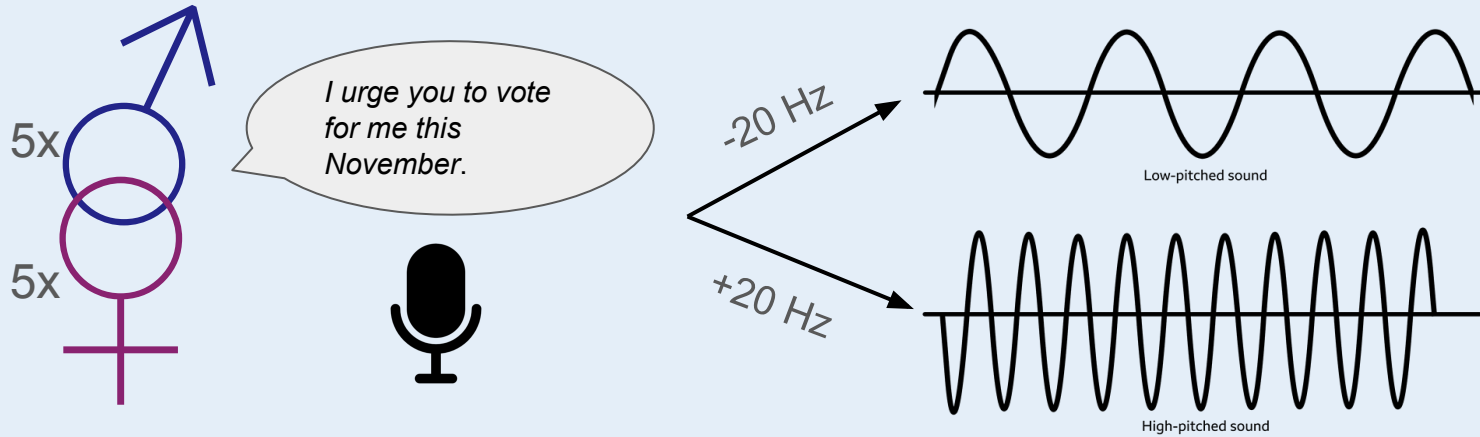
The Importance of Voice Pitch



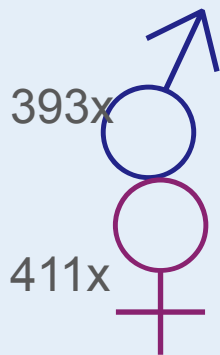
https://commons.wikimedia.org/wiki/File:Roaring_lion_%28Unsplash%29.jpg



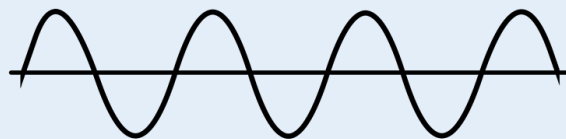
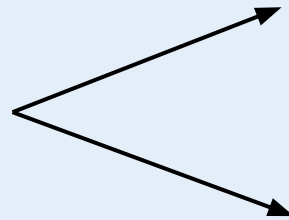
First Experiment - Setup



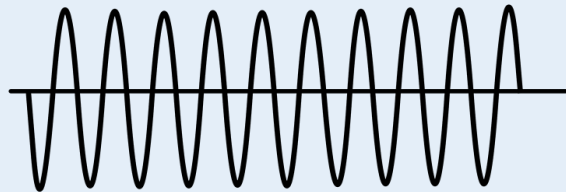
First Experiment - Setup



Vote 👍 👎



Low-pitched sound



High-pitched sound

Of different

- Age
- Race
- Education
- Political view
- Religion
- ...

5x same sex
speakers

First experiment - Results

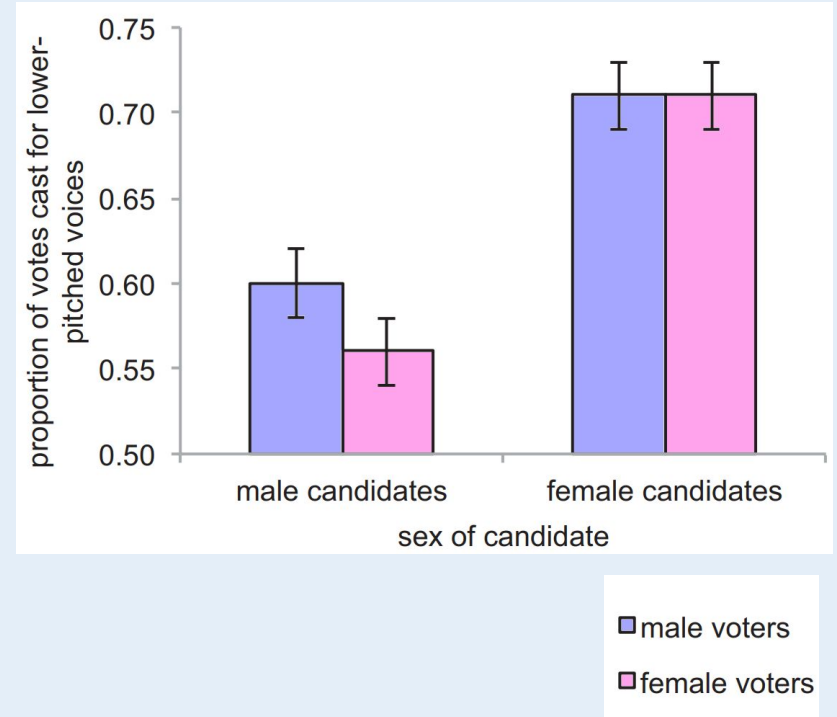
4 ANalysis Of VAriance (ANOVA) tests:

First experiment - Results

4 ANalysis Of VAriance (ANOVA) tests:

1. Voter vs candidate sex

- Both preferred low pitch, especially when voting female candidates



First experiment - Results

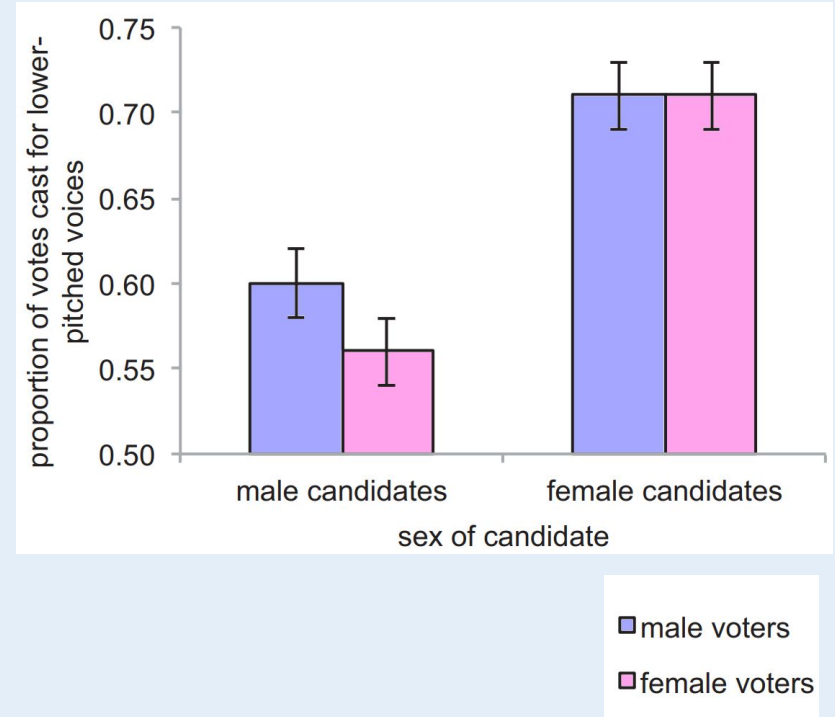
4 ANalysis Of VAriance (ANOVA) tests:

1. Voter vs candidate sex

- Both preferred low pitch, especially when voting female candidates

2. Voter age vs candidate sex

- Older voter prefer lower pitched



First experiment - Results

4 ANalysis Of VAriance (ANOVA) tests:

1. Voter vs candidate sex

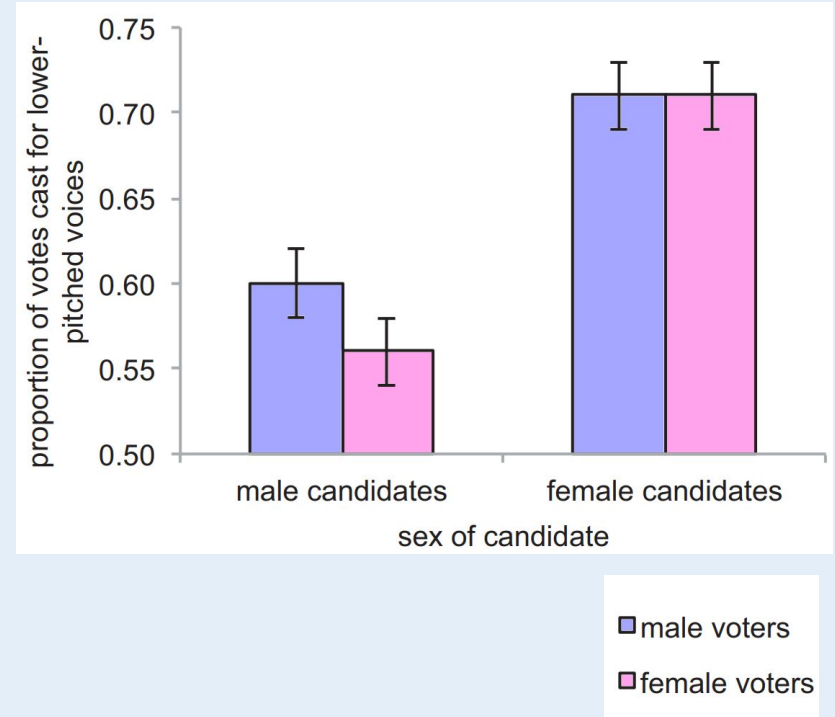
- Both preferred low pitch, especially when voting female candidates

2. Voter age vs candidate sex

- Older voter prefer lower pitched

3. Voter education vs candidate sex

- Voter with degree prefer lower pitched



First experiment - Results

4 ANalysis Of VAriance (ANOVA) tests:

1. Voter vs candidate sex

- Both preferred low pitch, especially when voting female candidates

2. Voter age vs candidate sex

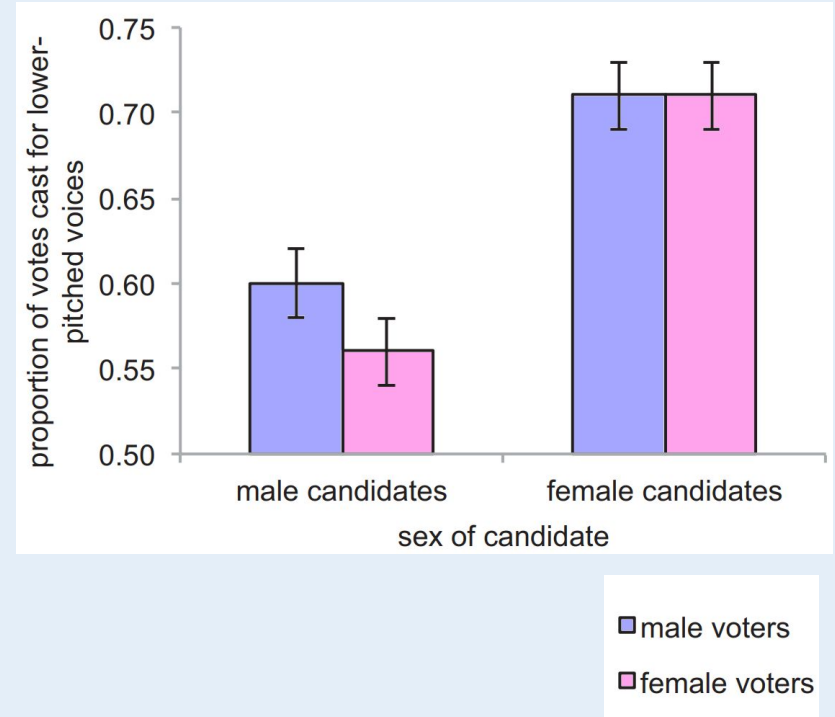
- Older voter prefer lower pitched

3. Voter education vs candidate sex

- Voter with degree prefer lower pitched

4. Political engagement vs candidate sex

- Highly engaged prefer lower pitched

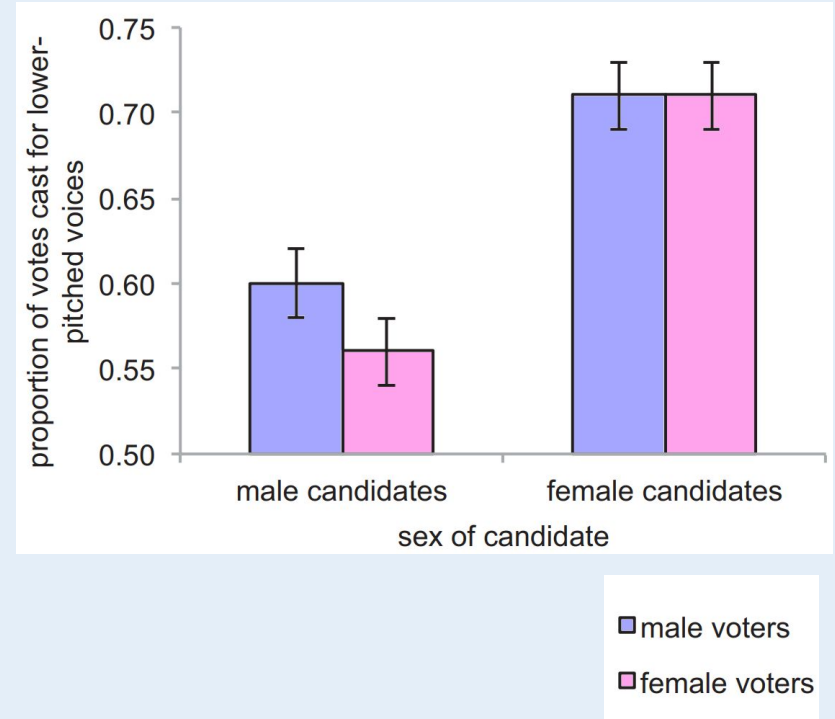


First experiment - Results

4 ANalysis Of VAriance (ANOVA) tests:

1. Voter vs candidate sex
 - Both preferred low pitch, especially when voting female candidates
2. Voter age vs candidate sex
 - Older voter prefer lower pitched
3. Voter education vs candidate sex
 - Voter with degree prefer lower pitched
4. Political engagement vs candidate sex
 - Highly engaged prefer lower pitched

Interactions not statistically significant



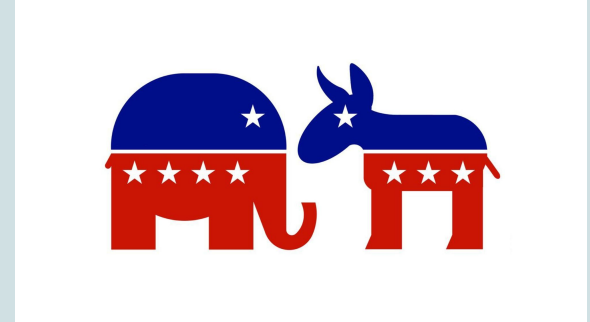
Second study - Setup

- 2012 U.S. House of Representatives
 - Per district, 2 with largest vote share
 - Recordings of voice from YouTube videos
 - In total 796 candidates (160 women, 636 men)
- Measured the pitch of recorded voices
 - Mean in Hz across audio file
 - For 722 candidates it was also possible to compare to the opponent
- Two election variables
 - Candidate won/lost (outcome)
 - Share of votes



Second study - Method of Analysis

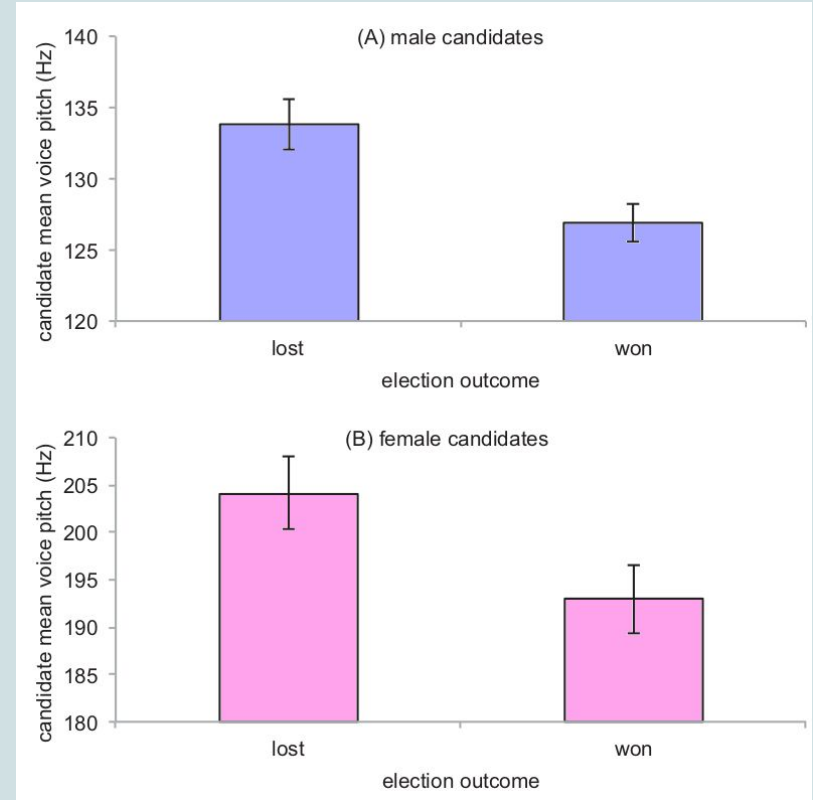
- Two-tailed two-sample t-tests for whether the election was won or lost and Pearson correlation for vote share.
- Regression on outcome and vote share using following features
 - Pitch
 - Sex
 - Sex * Pitch
 - Democrat/Republican
 - Barack Obama's or Mitt Romney's vote share for resp. Democrat or Republican candidate in their district
 - Natural log of candidate's total campaign spendings
 - Candidate has held elected office in the past/is currently an incumbent



https://media.cnn.com/api/v1/images/stellar/prod/181105112842-donkey-elephant-top.jpg?q=w_2000,c_fill/f_webp

Results - Candidate Mean Voice Pitch

- Higher for the ones who lost, both for female and male candidates
- Negative correlation between mean voice pitch and vote share for both sexes
- Concerning regression
 - Candidates with lower voices were more likely to win and get a larger vote share against male opponents
 - This did not vary significantly by the sex of the candidate
 - Opposite by female opponent (the higher, the larger the chance)
 - The relationship is stronger by male candidates



Results - Candidate Has Lower Voice than Opponent

- Male candidates with lower voices than opponents were more likely to win
- Same with female, but not statistically significant (perhaps smaller sample)
- The same can be said regarding the vote share
- Regarding regression
 - No strongly significant pattern, but suggests that candidates who had a lower voice than male opponents tended to win more share of votes

Discussion

- Candidate voice pitch can influence electoral outcomes
 - This is more observable in the first experiment
 - In the second one more variables come into play
- First time that was shown a case where candidates with higher pitch are preferred (male competing against female)
 - Male perceived as aggressive?
- Older, well-educated, and politically engaged voters are the most biased
 - Also the ones that most likely to turn out to vote!
- Female have higher-pitched voice
 - Contribute to gender inequality?

Questions for the Class Discussion

1. Is pitch the only feature of voice that could influence the voter's perception of the candidate? What else could be considered? Would it be easy to measure? In which way you predict it can influence the perception?
2. How do you think could be explained the different pattern observed when male candidates are competing against female candidates? Do you think the low-pitched male voice could be perceived as aggressive as suggested by the author?

Sources

- Tusing, K. J., & Dillard, J. P. (2000). The sounds of dominance: Vocal precursors of perceived dominance during interpersonal influence. *Human Communication Research*, 26, 148–171.
- Titze, I. R. (1994). *Principles of voice production*. Engelwood Cliffs, NJ: Prentice Hall.
- Baken, R. J. (1987). *Clinical measurement of speech and voice*. London, UK: Taylor and Francis.
- Feinberg, D. R., DeBruine, L. M., Jones, B. C., & Perrett, D. I. (2008). The role of femininity and averageness of voice pitch in aesthetic judgments of women's voices. *Perception*, 37, 615–623.
- Puts, D. A., Hodges, C., Cardenas, R. A., & Gaulin, S. J. C. (2007). Men's voices as dominance signals: Vocal fundamental and formant frequencies influence dominance attributions among men. *Evolution and Human Behavior*, 28, 340–344.
- Collins, S. A., & Missing, C. (2003). Vocal and visual attractiveness is related in women. *Animal Behaviour*, 65, 997–1004.
- Banse, R., & Scherer, K. R. (1996). Acoustic profiles in vocal emotion expression. *Journal of Personality and Social Psychology*, 70, 614–636.
- Kuroda, I., Fujiwara, O., Okamura, N., & Utsuki, N. (1976). Method for determining pilot stress through analysis of voice communication. *Aviation, Space, and Environmental Medicine*, 47, 528–533.
- Gregory, S. W., & Gallagher, T. J. (2002). Spectral analysis of candidates' nonverbal vocal communication: Predicting U.S. presidential election outcomes. *Sociology and Psychology Quarterly*, 65, 298–308.
- Tigue, C. C., Borak, D. J., O'Connor, J. J. M., Schandl, C., & Feinberg, D. R. (2012). Voice pitch influences voting behaviour. *Evolution and Human Behavior*, 33, 210–216.
- Klostad, C. A. (2011). *Civic talk: peers, politics, and the future of democracy*. Philadelphia, PA: Temple University Press.
- Klostad, C. A., Anderson, R. C., & Peters, S. (2012). Sounds like a winner: Voice pitch influences perception of leadership capacity in both men and women. *Proceedings of the Royal Society of London B*, 297, 2698–2704.

EXTRA SLIDES

The Two Studies

1. Experiment in a “controlled environment”
2. Outcomes in real elections

First Experiment - setup

- *I urge you to vote for me this November*
- Recording manipulated to have a higher and lower pitch version (roughly $\pm 20\text{Hz}$)
- 5 women aged 21-38 and 5 men aged 20-41
- Ev. insert plot of frequency distr.
- 393 men + 411 women as *electors*
- sex, age, race, years of education, interest in politics, employment status, Evangelical or born again Christian status, marital status, partisanship, and ideology follow a target sample based on U.S. Citizens demographics.
- Listened randomly either 5 pairs (higher vs lower) of male or 5 pairs of male

Table 1. Relationship between candidate mean voice pitch and outcomes of 2012 U.S. House elections

	Opponent is Male		Opponent is Female	
	Candidate Won ^a	Candidate Vote Share ^b	Candidate Won ^a	Candidate Vote Share ^b
Candidate mean voice pitch	-.01*	-.03*	.03*	-.005
	(.007)	(.01)	(.01)	(.02)
Candidate sex (female)	-2.97	-6.56	8.90**	2.84
	(2.63)	(4.13)	(3.30)	(10.25)
Candidate sex* Candidate mean voice pitch	.02	.04	-.06***	-.02
	(.02)	(.02)	(.02)	(.06)
Candidate party affiliation (Democrat)	-.38	-.04	-.75	-1.19
	(.35)	(.70)	(1.08)	(1.34)
Copartisan presidential candidate district vote share	.21***	.82***	.23***	.75***
	(.03)	(.04)	(.07)	(.09)
Natural log of campaign spending	.84***	.48***	1.37**	1.06***
	(.17)	(.10)	(.44)	(.32)
Quality candidate (has held elected office in past)	1.77***	7.13***	-.03	4.14*
	(.30)	(.93)	(1.00)	(1.68)
Constant	-21.08***	2.70	-33.48***	-2.85
	(3.25)	(2.17)	(6.46)	(4.66)
χ^2	90.55***	—	50.10***	—
F	—	524.06***	—	112.70***
Pseudo R ²	.68	.83	.66	.82
N	616	616	152	152

Note. Cell entries are regression coefficients. Cases with missing data were omitted from the analysis using listwise deletion. Model type: ^alogistic regression; ^blinear regression.

* $p \leq .05$; ** $p \leq .01$; *** $p \leq .001$ (robust standard errors, clustered by House district, in parentheses).

	Opponent is Male		Opponent is Female	
	Candidate Won ^a	Candidate Vote Share ^b	Candidate Won ^a	Candidate Vote Share ^b
Candidate mean voice pitch	-.01*	-.03*	.03*	-.005
	(.007)	(.01)	(.01)	(.02)
Candidate sex (female)	-2.97	-6.56	8.90**	2.84
	(2.63)	(4.13)	(3.30)	(10.25)
Candidate sex* Candidate mean voice pitch	.02	.04	-.06***	-.02
	(.02)	(.02)	(.02)	(.06)

Table 2. Relationship between Candidate Having a Lower Voice than Opponent and Outcomes of 2012 U.S. House Elections

	Opponent Is Male		Opponent Is Female	
	Candidate Won ^a	Candidate Vote Share ^b	Candidate Won ^a	Candidate Vote Share ^b
Candidate has lower voice	.16 (.41)	1.19 (.72)	.32 (.78)	-1.32 (1.53)
Candidate sex (female)	-.01 (.46)	.53 (.80)	-1.56 (1.15)	-3.50 (3.43)
Candidate sex* Candidate has lower voice	.69 (.82)	.02 (2.68)	1.28 (1.41)	2.11 (2.90)
Candidate party affiliation (Democrat)	-.44 (.37)	-.64 (.72)	-.41 (1.12)	-1.44 (1.41)
Copartisan presidential candidate district vote share	.23*** (.03)	.81*** (.04)	.21** (.07)	.74*** (.09)
Natural log of campaign spending	.89*** (.20)	.51*** (.10)	1.32*** (.47)	1.09*** (.33)
Quality candidate (has held elected office in past)	1.62*** (.31)	6.43*** (.91)	.08 (.91)	4.21* (1.78)
Constant	-24.76*** (3.82)	-1.24 (1.24)	-28.74*** (6.16)	-2.02 (4.11)
χ^2	65.19***	—	39.84***	—
F	—	391.20***	—	98.99***
Pseudo R ²	.68	.67	.63	.81
N	561	561	146	146

Note. Cell entries are regression coefficients. Cases with missing data were omitted from the analysis using listwise deletion. Model type: ^alogistic regression; ^blinear regression.

* $p \leq .05$; ** $p \leq .01$; *** $p \leq .001$ (robust standard errors, clustered by House district, in parentheses).

	Opponent Is Male		Opponent Is Female	
	Candidate Won ^a	Candidate Vote Share ^b	Candidate Won ^a	Candidate Vote Share ^b
Candidate has lower voice	.16 (.41)	1.19 (.72)	.32 (.78)	-1.32 (1.53)
Candidate sex (female)	-.01 (.46)	.53 (.80)	-1.56 (1.15)	-3.50 (3.43)
Candidate sex* Candidate has lower voice	.69 (.82)	.02 (2.68)	1.28 (1.41)	2.11 (2.90)

Conclusion

- Electoral outcomes showed subtler relationship with acoustic qualities when considering other candidates' features
- First study to show that some candidates with higher-pitched voices are more successful (males vs females)
- Older, well-educated, politically engaged voters are the most biased in favor of candidates with lower voices
 - Also the most likely to turn out to vote
- The bias in favor of candidates with lower voices is the strongest for female candidates
-